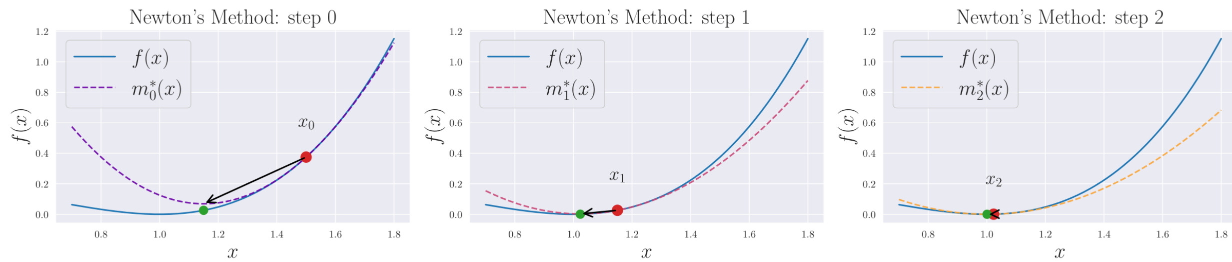

Figure 1: Illustration of iterations of Newton's method. In this example, $\alpha_k = 1$ holds for all steps, i.e., $x_{k+1} = x_k - \nabla^2 f(x_k)^{-1} g_k$.

1 Newton's Method

We define $f\colon \mathbb{R}^n \to \mathbb{R}$ as an objective function of class C^2, where n denotes the dimension of the decision variables. In this context, the following unconstrained optimization problem is one of the most fundamental optimization problems:

$$\underset{x\in\mathbb{R}^n}{\text{minimize}} \quad f(x).$$

In this section, we outline Newton's method, which is a fundamental algorithm for this unconstrained optimization problem.

1.1 Algorithm of Newton's Method

Newton's method is a representative iterative algorithm that starts from an initial point x_0, successively updates the current point, and generates a sequence $\{x_k\}_k$. We assume that f is of class C^2 and strongly convex, so that the Hessian matrix $\nabla^2 f(x)$ is positive definite for all $x \in \mathbb{R}^n$, and thus invertible. Using the gradient $g_k := \nabla f(x_k)$ and Hessian $\nabla^2 f(x_k)$ at the k-th iterate x_k, the one step of Newton's method can be expressed as follows:

$$x_{k+1} \leftarrow x_k - \nabla^2 f(x_k)^{-1} g_k.$$

We can derive this update rule as follows. Firstly, the quadratic model of f at x_k is given by the Taylor expansion as follows:

$$m_k^*(x) := f(x_k) + g_k^\top (x - x_k) + \frac{1}{2}(x - x_k)^\top \nabla^2 f(x_k)(x - x_k).$$

The gradient of this model is

$$\nabla m_k^*(x) = g_k + \nabla^2 f(x_k)(x - x_k).$$

Since the quadratic model is strongly convex by assumption, a point $x^* \in \mathbb{R}^n$ is the minimizer of m_k^* if and only if $\nabla m_k^*(x) = 0$. Thus, solving this equation yields

$$\underset{x\in\mathbb{R}^n}{\arg\min}\, m_k^*(x) = x_k - \left(\nabla^2 f(x_k)\right)^{-1} g_k.$$

Thus, it is natural to select this x^* as the next iterate, and this leads to the update rule of Newton's method.

However, as we will see later, it does not generally guarantee global convergence. Thus, we often introduce a step size $\alpha_k > 0$ determined by line search to ensure a sufficient decrease in function values, leading to the update rule as follows:

$$x_{k+1} \leftarrow x_k - \alpha_k \nabla^2 f(x_k)^{-1} g_k.$$

This variant is called the damped Newton's method or relaxed Newton's method, and it is a common practice to use this version in general optimization problems.

The update direction $d_k := -\nabla^2 f(x_k)^{-1} g_k$ is known as the Newton direction. When the Hessian is positive definite, the following relation shows that it is a descent direction, meaning that it is a direction in which the objective function decreases.

$$g_k^\top d_k = -g_k^\top \nabla^2 f(x_k)^{-1} g_k < 0.$$

When f is nonconvex, meaning that the Hessian $\nabla^2 f(x_k)$ is negative definite or indefinite, d_k may not be a descent direction, and Newton's method may fail to converge. The modified Newton's method [2, Sec. 3.4] is one remedy for handling such cases, but we omit the explanation here.

1.2 Convergence Properties of Newton's Method

We next present standard convergence properties for Newton's method.

1.2.1 Global Convergence

We first state a general global convergence result

$$\lim_{k\to\infty} \|g_k\| = 0$$

for methods with line search, and then explain its application to damped Newton's method.

We consider a general iterative method of the form

$$x_{k+1} \leftarrow x_k + \alpha_k d_k.$$

The direction d_k is a descent direction satisfying $g_k^\top d_k < 0$ and $\alpha_k > 0$ is the step size determined by line search. We employ the (weak) Wolfe conditions [2, Sec. 3.1] to determine the step size α_k as follows:

$$\begin{cases} f(x_k + \alpha_k d_k) & \leq f(x_k) + c_1 \alpha_k g_k^\top d_k, \\ -g_{k+1}^\top d_k & \leq -c_2 g_k^\top d_k. \end{cases}$$

$0 < c_1 < c_2 < 1$ are constants.

The first condition is known as the Armijo condition, which requires that the function value at the next point $f(x_k + \alpha_k d_k)$ decreases by at least a fraction c_1 of the predicted decrease $\alpha_k g_k^\top d_k$.

The second condition is known as the curvature condition. If d_k is a descent direction at the next point x_{k+1}, then $-g_{k+1}^\top d_k$ is positive, and the objective function value decreases if we move slightly in the direction of d_k. The curvature condition requires that this decrease rate is not too large, meaning that a sufficiently large step size is taken along d_k. In the weak Wolfe conditions, it is also possible that the step size is too long, but in the strong Wolfe conditions, by taking the absolute value, it requires that an appropriate step size is taken.

The Wolfe conditions refer to the simultaneous satisfaction of Armijo and curvature conditions. Figure 2 illustrates the ranges where these conditions are satisfied.

We also define $\theta_k \in [0, \pi]$ as the angle between the direction d_k and the negative gradient $-g_k$, satisfying

$$\cos\theta_k = \frac{-g_k^\top d_k}{\|g_k\| \, \|d_k\|}.$$

Under these settings, we present a simplified version of the following classical well-known result.

Theorem 1 ([2, Theorem 3.2]). *Suppose that $f\colon \mathbb{R}^n \to \mathbb{R}$ is of class C^1 and bounded below in $\mathbb{R}^n$, and L-smooth. Consider the following iterative method*

$$x_{k+1} \leftarrow x_k + \alpha_k d_k.$$

It starts from an initial point $x_0 \in \mathbb{R}^n$, and the step size α_k satisfies the Wolfe conditions. For the angles $\{\theta_k\}_k$, if there exists a positive constant δ such that $\cos\theta_k \geq \delta > 0$ for all k, then the method generates a sequence $\{x_k\}_k$ satisfying

$$\lim_{k\to\infty} \|g_k\| = 0.$$

Proof. From the Wolfe conditions, Cauchy–Schwarz inequality and the L-smoothness of f, the following two relations hold:

$$\begin{cases} (g_{k+1} - g_k)^\top d_k = g_{k+1}^\top d_k - g_k^\top d_k \geq (c_2 - 1) g_k^\top d_k, \\ (g_{k+1} - g_k)^\top d_k \leq \|g_{k+1} - g_k\| \, \|d_k\| \leq L \|x_{k+1} - x_k\| \, \|d_k\| = \alpha_k L \|d_k\|^2. \end{cases}$$

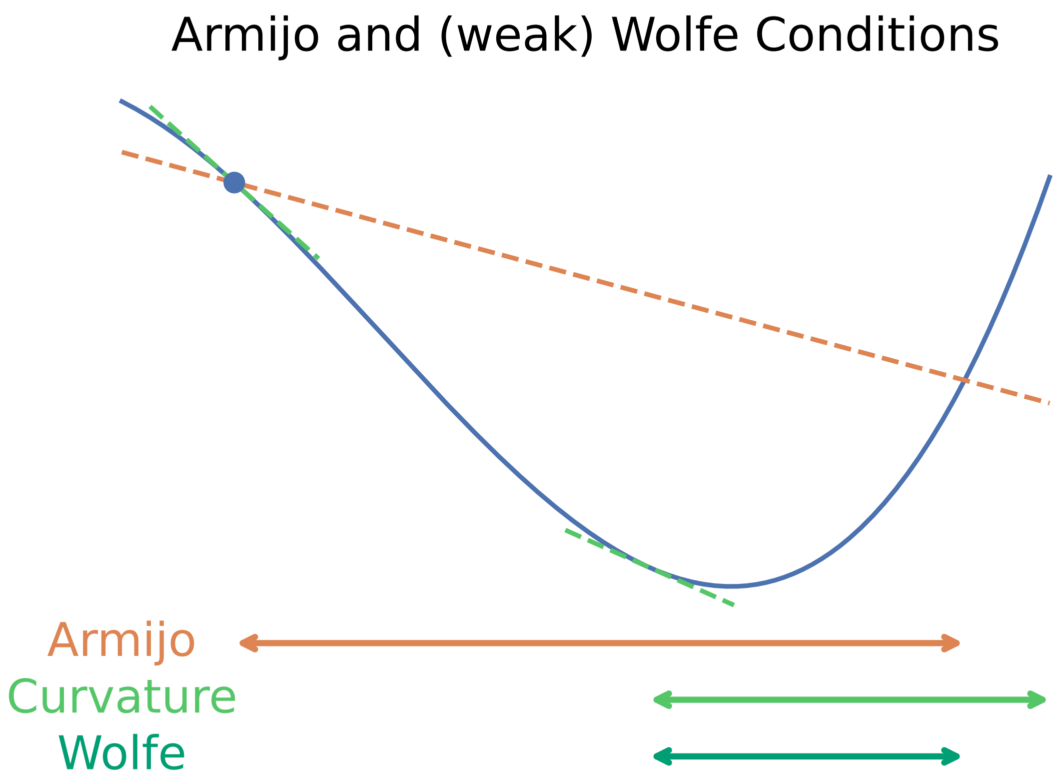

Figure 2: Illustration of the line search conditions. Each arrow indicates the range where the corresponding condition is satisfied.

By combining these two relations, we obtain

$$\alpha_k \geq \frac{c_2 - 1}{L} \frac{g_k^\top d_k}{\|d_k\|^2}.$$

Thus, since $\cos\theta_k > 0$ implies that d_k is a descent direction, i.e., $g_k^\top d_k < 0$, we have the following inequality.

$$\begin{aligned}
f(x_{k+1}) &\leq f(x_k) + c_1 \alpha_k g_k^\top d_k && \text{(Armijo condition)} \\
&\leq f(x_k) - c_1 \frac{1-c_2}{L} \frac{(g_k^\top d_k)^2}{\|d_k\|^2} && \text{(previous inequality)} \\
&= f(x_k) - c_1 \frac{1-c_2}{L} \cos^2\theta_k \|g_k\|^2. && \text{(definition of } \theta_k\text{)}
\end{aligned}$$

By repeatedly applying this inequality, we obtain

$$f(x_{k+1}) \leq f(x_0) - c_1 \frac{1-c_2}{L} \sum_{j=0}^{k} \cos^2\theta_j \|g_j\|^2.$$

Since we assumed that f is bounded below, there exists a constant f^* such that $f(x_{k+1}) \geq f^*$ for all k. Hence, we obtain the following condition known as the Zoutendijk condition:

$$\sum_{k=0}^{\infty} \cos^2\theta_k \|g_k\|^2 \leq \frac{L}{c_1(1-c_2)} (f(x_0) - f^*) < \infty.$$

This condition implies that

$$\cos^2\theta_k \|g_k\|^2 \to 0.$$

Combined with the assumption that $\cos\theta_k \geq \delta > 0$ for all k, it follows immediately that

$$\lim_{k\to\infty} \|g_k\| = 0.$$

This completes the proof. $\square$

Since damped Newton's method is a special case of the setting in Theorem 1 with $d_k = -\nabla^2 f(x_k)^{-1} g_k$, we can apply this result under appropriate conditions of f. In particular, if f is L-smooth and μ-strongly convex, then for any k, we have

$$\cos\theta_k = \frac{g_k^\top \nabla^2 f(x_k)^{-1} g_k}{\|g_k\| \, \|\nabla^2 f(x_k)^{-1} g_k\|} \geq \frac{\|g_k\|^2 \lambda_{\min}(\nabla^2 f(x_k)^{-1})}{\|g_k\|^2 \lambda_{\max}(\nabla^2 f(x_k)^{-1})} \geq \frac{\mu}{L}.$$

Here, $\lambda_{\min}(\cdot)$ and $\lambda_{\max}(\cdot)$ denote the minimum and maximum eigenvalues, respectively. Thus, by setting $\delta = \mu/L$ in Theorem 1, we obtain the global convergence of damped Newton's method for strongly convex and smooth functions under line search satisfying the Wolfe conditions.

1.2.2 Local Quadratic Convergence

We next present a classical result on the local convergence of Newton's method. We again provide a simplified version for brevity.

Theorem 2 ([2, Theorem 3.5])**.** *Suppose that the Hessian matrix $\nabla^2 f(x)$ is Lipschitz continuous in a neighborhood of the solution x^*, and that sufficient second-order optimality conditions hold (i.e., $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*)$ is positive definite). In Newton's method, if $\alpha_k = 1$ holds for all k and the initial point x_0 is sufficiently close to x^*, then the sequence of gradient norms $\{\|\nabla f(x_k)\|\}_k$ converges quadratically.*

Proof. Let $k = 0$ and consider the update from x_k to x_{k+1}. Since the Hessian $\nabla^2 f(x_k)$ is invertible in a sufficiently small neighborhood of x^*, from the optimality condition $\nabla f(x^*) = 0$ and the assumption $\alpha_k = 1$, we have

$$\begin{aligned}
x_{k+1} - x^* &= x_k - x^* - \left(\nabla^2 f(x_k)\right)^{-1} \nabla f(x_k) \\
&= \left(\nabla^2 f(x_k)\right)^{-1} \left(\nabla^2 f(x_k)(x_k - x^*) - (\nabla f(x_k) - \nabla f(x^*))\right).
\end{aligned}$$

By Taylor's theorem and the triangle inequality, we have

$$\begin{aligned}
&\left\|\nabla^2 f(x_k)(x_k - x^*) - (\nabla f(x_k) - \nabla f(x^*))\right\| \\
&= \left\|\nabla^2 f(x_k)(x_k - x^*) - \int_0^1 \nabla^2 f(x_k + t(x^* - x_k))(x_k - x^*)\mathrm{d}t\right\| \\
&\leq \int_0^1 \left\|\nabla^2 f(x_k) - \nabla^2 f(x_k + t(x^* - x_k))\right\| \|x_k - x^*\| \,\mathrm{d}t.
\end{aligned}$$

If $\nabla^2 f$ is Lipschitz continuous with constant L^{H}, then the integrand is bounded by $L^{\mathrm{H}} t \|x_k - x^*\|$, and integrating yields

$$\left\|\nabla^2 f(x_k)(x_k - x^*) - (\nabla f(x_k) - \nabla f(x^*))\right\| \leq \frac{1}{2} L^{\mathrm{H}} \|x_k - x^*\|^2.$$

Since $\nabla^2 f(x^*)$ is positive definite, there exists a radius $r > 0$ such that for all x satisfying $\|x - x^*\| \leq r$,

$$\left\|\left(\nabla^2 f(x)\right)^{-1}\right\| \leq 2\left\|\left(\nabla^2 f(x^*)\right)^{-1}\right\|$$

holds. Combining these results, we obtain

$$\begin{aligned}
\|x_{k+1} - x^*\| &\leq \left\|\left(\nabla^2 f(x_k)\right)^{-1}\right\| \left\|\nabla^2 f(x_k)(x_k - x^*) - (\nabla f(x_k) - \nabla f(x^*))\right\| \\
&\leq L^{\mathrm{H}} \left\|\left(\nabla^2 f(x^*)\right)^{-1}\right\| \|x_k - x^*\|^2.
\end{aligned}$$

Let $\widetilde{L} := L^{\mathrm{H}} \left\| \left(\nabla^2 f(x^*)\right)^{-1} \right\|$. If the initial point is chosen such that $\|x_0 - x^*\| \le \min\{r, 1/(2\widetilde{L})\}$, then by induction $\{x_k\}_k$ remains within the neighborhood and converges to x^*. Thus, the above error bound implies quadratic convergence of $\{x_k\}_k$.

To show quadratic convergence of the gradient norm, we use $x_{k+1} - x_k = -\left(\nabla^2 f(x_k)\right)^{-1} \nabla f(x_k)$ and $\nabla f(x_k) + \nabla^2 f(x_k)(x_{k+1} - x_k) = 0$, yielding

$$\begin{aligned}
\|\nabla f(x_{k+1})\| &= \left\|\nabla f(x_{k+1}) - \nabla f(x_k) - \nabla^2 f(x_k)(x_{k+1} - x_k)\right\| \\
&\le \int_0^1 \left\|\nabla^2 f(x_k + t(x_{k+1} - x_k)) - \nabla^2 f(x_k)\right\| \|x_{k+1} - x_k\| \, \mathrm{d}t \\
&\le \frac{1}{2} L^{\mathrm{H}} \|x_{k+1} - x_k\|^2 \\
&\le \frac{1}{2} L^{\mathrm{H}} \left\| \left(\nabla^2 f(x_k)\right)^{-1} \right\|^2 \|\nabla f(x_k)\|^2 \\
&\le 2L^{\mathrm{H}} \left\| \left(\nabla^2 f(x^*)\right)^{-1} \right\|^2 \|\nabla f(x_k)\|^2 .
\end{aligned}$$

Therefore, $\|\nabla f(x_k)\|$ converges quadratically to 0. □

These Theorems 1 and 2 indicate that Newton's method has both global convergence and local quadratic convergence properties under appropriate conditions. This rapid convergence is a significant advantage of Newton's method compared to first-order methods such as gradient descent, which typically exhibit only linear convergence.

1.3 Elements for Global Convergence

As mentioned earlier, the positive definiteness of the Hessian matrix and the selection of the step size via line search play important roles in Newton's method. In this subsection, we explain why these elements are required.

1.3.1 Positive Definiteness of the Hessian Matrix

We have assumed that f is strongly convex so far, meaning that the Hessian matrix $\nabla^2 f(x_k)$ is positive definite at each iteration. This assumption is crucial for Newton's method to converge locally to an optimal solution. When the Hessian matrix is positive definite, Newton's method provides a descent direction, since

$$\nabla f(x_k)^\top d_k = -\nabla f(x_k)^\top \left(\nabla^2 f(x_k)\right)^{-1} \nabla f(x_k) < 0.$$

In contrast, when the Hessian matrix is negative definite or indefinite, a decrease in the function value is not guaranteed, and Newton's method can move away from the optimal solution. In the indefinite case, the function value may decrease, but the risk of converging to a saddle point is also increased. Therefore, the positive definiteness of the Hessian matrix is important when applying Newton's method.

1.3.2 Issues with Full Steps

Newton's method with $\alpha_k = 1$ at every iteration may lead to problematic behavior. Here, we present specific examples of such failures and discuss the necessity of line search as a countermeasure.

An Example Where Newton's Method Diverges Consider the following function:

$$\begin{aligned}
f(x) &:= \sqrt{1 + x^2}, \\
f'(x) &= \frac{x}{\sqrt{1 + x^2}}, \\
f''(x) &= \frac{1}{(1 + x^2)^{3/2}}.
\end{aligned}$$

When the absolute value of the initial point exceeds 1, Newton's method diverges for this function, as shown in Figure 3.

This is because, at points far from the optimal solution $x^* = 0$, the value of the second derivative, i.e., the eigenvalue of the Hessian matrix, becomes very small, leading to an excessively large Newton step.

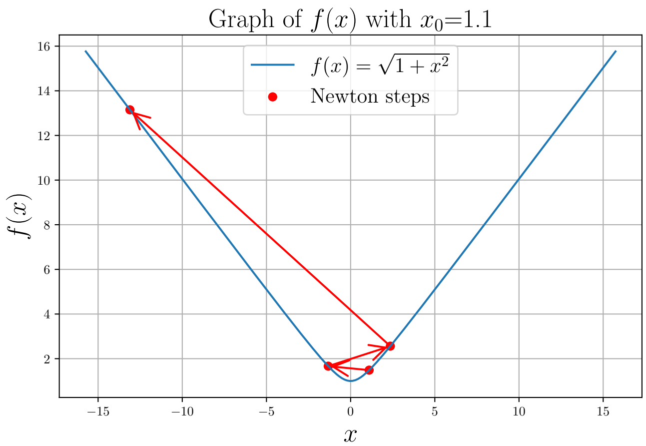

Figure 3: An example where Newton's method diverges with initial point $x_0 = 1.1$

Oscillation of Newton's Method for Strongly Convex Functions In the previous example, the objective function was not strongly convex and did not necessarily possess favorable properties. However, even for functions with the strong convexity property, there exist examples where Newton's method fails to converge [1, Example 1.4.3].

In this example, for $\mu > 0$, consider the function

$$\begin{aligned} f(x) &:= \log(1+e^x) - \frac{x}{2} + \frac{\mu x^2}{2}, \\ f'(x) &= \frac{e^x}{1+e^x} - \frac{1}{2} + \mu x, \\ f''(x) &= \frac{e^x}{(1+e^x)^2} + \mu, \\ f'''(x) &= \frac{e^x(1-e^x)}{(1+e^x)^3}, \\ f''''(x) &= \frac{e^x(1-4e^x+e^{2x})}{(1+e^x)^4}. \end{aligned}$$

This function has the following properties:

- It is μ-strongly convex.
- $\max_x |f''(x)| = \frac{1}{4} + \mu$ (attained at $e^x = 1$). Hence, ∇f is L-smooth with $L = \frac{1}{4} + \mu$.
- $\max_x |f'''(x)| = \frac{1}{6\sqrt{3}}$ (attained at $e^x = 2 - \sqrt{3}$). Hence, $\nabla^2 f$ is M-Lipschitz with $M = \frac{1}{6\sqrt{3}}$.

Nevertheless, when the initial point x_0 is sufficiently large relative to μ, Newton's method exhibits oscillatory behavior, as shown in Figure 4.

Necessity of Line Search From these examples, it is clear that adopting a full step size $\alpha_k = 1$ in Newton's method can lead to divergence or oscillation, even for strongly convex and smooth functions. Therefore, it is essential to select the step size α_k appropriately to ensure convergence. This is why damped Newton's method with line search is often used.

1.4 Comparison with Newton's Method as a Root-Finding Algorithm

As a conceptual supplement, we briefly clarify the relationship between "Newton's method in optimization" and "Newton's method as a root-finding algorithm." These two formulations are closely related, but they have different perspectives and applications.

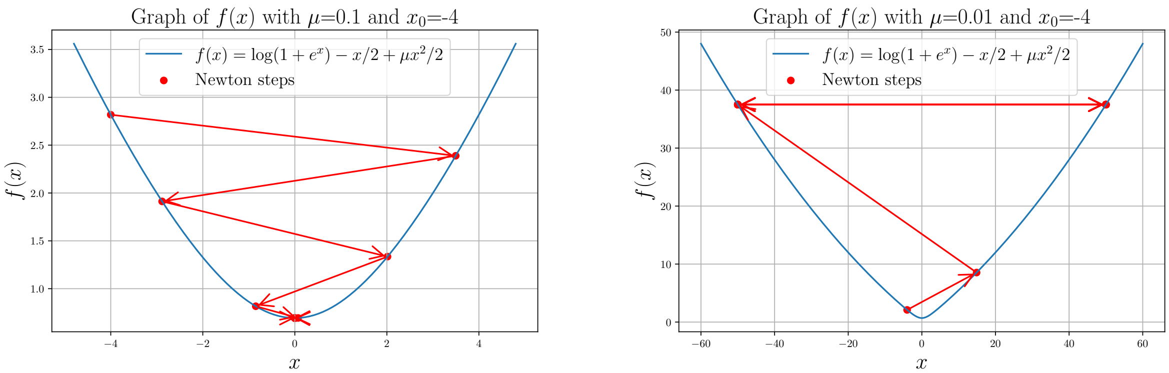

Figure 4: (Left) Newton's method converges for $x_0 = -4,\ \mu = 0.1$. (Right) Newton's method oscillates for $x_0 = -4,\ \mu = 0.01$.

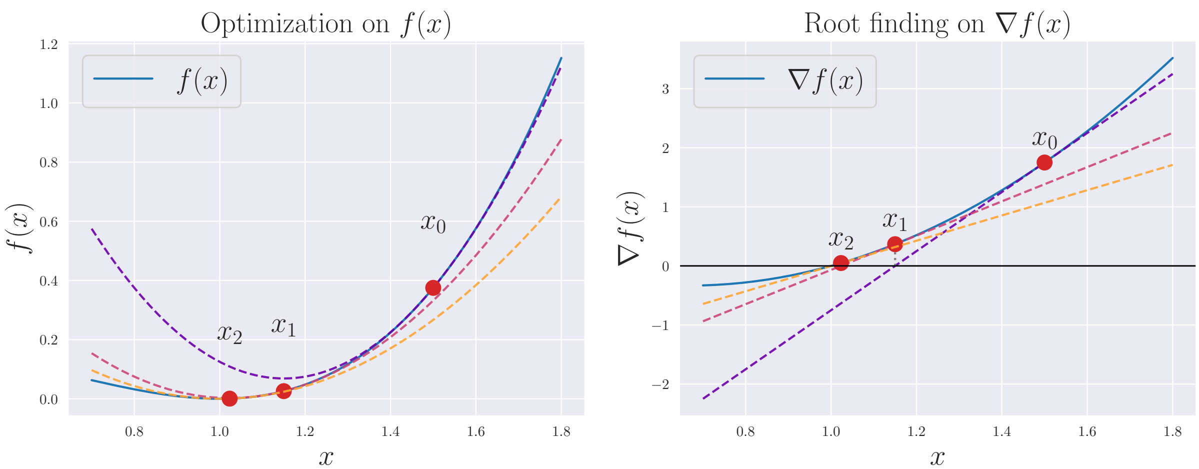

Figure 5: Optimization of the function $f(x) = x^3 - 2x^2 + x$ and root-finding for the gradient $\nabla f(x) = 3x^2 - 4x + 1$.

1.4.1 Newton's Method in Optimization

In the context of optimization, Newton's method typically refers to an algorithm for finding a local minimizer of a twice-differentiable function $f\colon \mathbb{R}^n \to \mathbb{R}$. The update of Newton's method with a step size $\alpha_k = 1$ was given by

$$x_{k+1} = x_k - \nabla^2 f(x_k)^{-1} \nabla f(x_k).$$

In the scalar case, the above expression reduces to $x_{k+1} = x_k - \frac{f'(x_k)}{f''(x_k)}$.

1.4.2 Newton's Method as a Root-Finding Algorithm

When referring to Newton's method (or the Newton–Raphson method), one may mean a root-finding algorithm for a differentiable scalar function $g\colon \mathbb{R} \to \mathbb{R}$ that solves $g(x) = 0$. It is also arguably fundamental and widely known.

Starting from an initial value x_0, the iteration is given by

$$x_{k+1} = x_k - \frac{g(x_k)}{g'(x_k)}.$$

Note that in the scalar case, since $g(x_k) = f'(x_k)$, this is the same formula as the update of Newton's method in optimization we saw earlier. Geometrically, this corresponds to approximating the graph of g near x_k by its tangent line and selecting the intersection of this tangent with the x-axis as the next approximate solution.

1.4.3 Relationship Between the Two Formulations

Let us examine the equivalence of the two formulations through a concrete example. Figure 5 illustrates the relationship between these formulations using a simple cubic function $f(x)$. The left figure corresponds to optimization of $f(x)$, while the right figure corresponds to root-finding for $g(x) = \nabla f(x)$. In the optimization formulation, $f(x) = x^3 - 2x^2 + x$ is approximated by its second-order Taylor expansion

$$m_k^*(x) = f(x_k) + \nabla f(x_k)(x - x_k) + \frac{1}{2}\nabla^2 f(x_k)(x - x_k)^2$$

and the minimizer of this quadratic model is taken as x_{k+1}. In the root-finding formulation, $\nabla f(x) = 3x^2 - 4x + 1$ is approximated by its tangent line, namely

$$\nabla m_k^*(x) = \nabla f(x_k) + \nabla^2 f(x_k)(x - x_k)$$

and the root of this linear model is chosen as x_{k+1}. Thus, under the assumption of positive definiteness, it is clear that both formulations perform essentially equivalent operations, and their correspondence can be explicitly verified.

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[2] Jorge Nocedal and Stephen J Wright. *Numerical Optimization*. Springer, 1999. ISBN 978-0-387-30303-1. doi: 10.1007/978-0-387-40065-5.